

Ali Asghar Al Qamari

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Professional Profile

Systems Engineer with 5+ years of experience in power electronics, electric drive systems, and industrial automation, with strong capabilities in data analysis, reporting, and workflow automation.

Experienced in working with technical and operational datasets using **Python**, **Power BI**, **SQL**, and **Excel** to analyze system performance, automate processes, and build dashboards for data-driven decision-making.

Combines system-level engineering expertise with data analytics to deliver efficient, reliable, and insight-driven solutions in complex technical environments.

Professional Experience

May 2023 – Present **Systems Engineer & Data Analytics – Power Electronics**, *Schaeffler AG*, Nuremberg, Germany

- Defined and managed system requirements for high-voltage inverter systems (up to 800V), ensuring alignment with technical and regulatory standards.
- Analyzed system performance using simulation data and technical datasets to evaluate efficiency and operational behavior.
- Applied **Python** and **Excel** for data processing, analysis, and automation of engineering workflows.
- Developed **Power BI dashboards** for system validation, performance monitoring, and reporting.
- Performed system-level validation using **MATLAB/Simulink** and **PLECS** (MiL/SiL/HiL).
- Collaborated with cross-functional teams to optimize system design using data-driven insights.
- Supported functional safety activities (ISO 26262) and system verification processes.

Feb 2022 – Apr 2023 **Master's Thesis & Internship – Power Electronics & System Analysis**, *Vitesco Technologies Germany GmbH*, Regensburg, Germany

- Analyzed inverter topologies and system performance using simulation data and technical datasets.
- Processed and evaluated simulation data using **Python (Pandas)** and **Excel** for performance comparison and trend analysis.
- Developed **Power BI dashboards** to visualize validation results and system performance metrics.
- Implemented control algorithms (SVPWM, FOC) using **MATLAB/Simulink**.
- Performed multi-domain simulations using **MATLAB/Simulink**, **PLECS**, **ANSYS Twin Builder**.
- Conducted thermal and electromagnetic analysis for system performance evaluation.
- Supported system validation and integration, including PLC-related control concepts.

- July 2018 – **Automation Engineer – Control Systems & Industrial Automation**, *Descon Engineering Limited*, Lahore, Pakistan
- September 2020
- Designed and implemented automation systems using **Siemens TIA Portal** with S7-1200/1500 PLCs.
 - Developed PLC logic and HMI/SCADA systems using **Siemens WinCC** for real-time monitoring and control.
 - Collected and analyzed operational data from industrial systems to monitor performance and improve efficiency.
 - Supported data visualization and reporting through SCADA dashboards and system monitoring tools.
 - Performed commissioning, testing, and troubleshooting for industrial automation systems.
 - Worked with industrial communication protocols (**PROFINET**, **Modbus**, **Ethernet**).

Projects

Power Electronics Performance Dashboard – Power BI, Python, Excel Interactive dashboard for analyzing inverter efficiency, losses, and system performance.

Simulation Data Analysis Tool – Python, Pandas, Matplotlib Automated workflow for cleaning, analyzing, and visualizing technical simulation data.

Industrial Automation Monitoring System – PLC, WinCC, Excel Monitoring and visualization of industrial processes and operational parameters.

Engineering Data Pipeline – Python, Pandas, Excel Reusable pipeline for cleaning, transforming, and reporting technical datasets.

Education

- Nov 2020 – **M.Sc. Electrical Engineering and Information Technology**, *Hochschule Darmstadt*, Darmstadt, Germany
- Apr 2023
- Focus: Power Electronics, Control Systems, Embedded Systems
 - CGPA: 1.6 (German Grade System)
- Sep 2014 – **B.E Electrical Engineering**, *Hamdard University*, Karachi, Pakistan
- Apr 2018
- CGPA: 1.3 (German Grade System)

Technical Skills

Data & Analytics	Python (Pandas, Matplotlib), Power BI, SQL, Excel
Simulation	MATLAB/Simulink, PLECS, ANSYS Twin Builder, PSCAD
Systems Engineering	System architecture, validation (MiL/SiL/HiL), ISO 26262
Automation	PLC (TIA Portal, S7-1200/1500), SCADA (WinCC)
Communication	PROFINET, Modbus, Ethernet

Language

- English:** Professional working proficiency (C1)
- German:** Intermediate working proficiency (B1/B2)